

The Index of First-Order Indexes

An admissible chart for indexes of the periodic elements, its eight lawful channels, and what four refusals establish

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Abstract. A first-order index here is a finite set of cells charting some body of atomic, nuclear or particle data, and its members must carry quantum numbers. Five closure operators -- order, algebra, geometry, information and statistics -- act on such a set, and which of them close it is a property of the set rather than of the operators. We show that only 8 of the 32 subsets of the five can occur, characterise them as the down-sets of a seven-relation law, and use the resulting triple (channel, height, width) as an admissible chart. Charting every index this project seats gives a second-order object of 23 vertices on 23 distinct cells. We prove that 2-determinacy is vacuous at arity 2 and derive from it that an arity-2 chart can never occupy the two lowest channels, which explains why one channel was the last to be reached; all eight are now occupied by charts of real data, so the bound of the first theorem is tight from nature and not only by construction. We give a ruling admitting overlapping charts when they carry different information, four grounds on which it is tested, and a census over 232 modules establishing that no unseated index exists in the tree. Of five adjudications reported here, four are refusals and one is a retraction of a seating this paper's own method had previously accepted. Finally we adjudicate the DEMAND -- the cells the seated indexes' own closure asks for and no member occupies. We prove that the demand invents no coordinate value, and that a bound monotone in the coordinates can never forbid a demanded cell, which empties at a stroke every hydrogenic and Pauli constraint one might bring to it; we prove that the quark model forbids no cell of the meson chart, and locate the cost in a coordinate that chart had refused for totality; and we exhibit the one bound in the tree that does forbid, which empties more than half of the largest prediction set in the register.

1. The object, and the criterion that bounds it

A first-order index here is a finite set of tuples -- cells -- obtained by charting some body of atomic, nuclear or particle data on a fixed list of coordinates. The subject began as the periodic elements and was widened, by ruling, to quantum objects generally; the criterion did not move, and it is narrow and enforced in code rather than in prose: A MEMBER OF SUCH AN INDEX MUST CARRY QUANTUM NUMBERS. An electron, a subshell, a transition, a spectroscopic term, a nuclide-charge state, a meson, a baryon, a fractional-quantum-Hall quasiparticle and a nuclear excited state all qualify; a file, a build snapshot or a document does not, and neither does a chemical bond, a binding energy or a scattering channel -- each of which was tested against the criterion and refused on its own measured ground.

The criterion is a function and not a paragraph, and the reason is historical. When it was a paragraph, thirteen filing-system indexes were seated as vertices -- mirrored files, conversations, archives, handoff documents -- beside indexes whose members are electrons. Every one of the disruptive vertices was repository metadata and not one was an element. The explosion in demand that followed was reported as a finding; it was contamination, and it is withdrawn.

23 indexes satisfy the criterion and are seated.

index	method	one member is	quantum numbers	cells	cell
fibred	FIBRATIO N	170 electrons, MADELUNG-PREDICTED	n, l, k	170	(3, 26, 17)
madelung	FIBRATIO N	the same 170 electrons, MADELUNG-PREDICTED	n+l, l, k	170	(7, 30, 12)
ions	TABLE	98 Lambda-8 transitions	charge, electron count	98	(0, 18, 16)
channels	TABLE	209 spectroscopic channel shapes	l, Pauli bound, multiplicity	209	(0, 16, 24)
laws	RESIDUAL	584 series against Rydberg-Ritz	n range, l, quantum defect	54	(0, 25, 7)
probability	TABLE	25 subshells	n, l	25	(2, 8, 9)

index	method	one member is	quantum numbers	cells	cell
inversion	TABLE	20 fill-order/shell-order inversions	pairs of (n, l)	17	(2, 6, 5)
gravity	TABLE	3,394 nuclide-charge states x 8 dimensions	Z, N, A, q, Ne, 2J, level status	914	(0, 19, 112)
nucshell	TABLE	22 nuclear single-particle subshells	nr, l, j (nuclear)	22	(3, 7, 6)
madrule	RESIDUAL	20 Madelung exceptions, by their transfer	n+l of the acceptor, l of the donor, occupancy	13	(2, 6, 4)
terms	TABLE	5,132 Russell-Saunders terms over 122 spectra	2S+1, L, parity, the banked J set	112	(0, 13, 18)
observed	FIBRATIO N	108 observed differentiating electrons, register 1306	n, l, k (observed)	98	(0, 20, 13)
gravity_bound	TABLE	the same 3,394 nuclide-charge states, on the bound structure alone	horizon-bound class, forced angular momentum, spin-decade rank	26	(1, 8, 5)
madelung_slot	FIBRATIO N	the same 170 electrons, subshell-blind	n+l, k	82	(6, 26, 6)
fundamental	TABLE	30 Standard Model particles -- 12 quarks, 12 leptons, 6 gauge/Higgs	2J, Q3, colour dimension, generation	26	(2, 7, 7)
mesons	TABLE	242 mesons of the PDG table, 8 set aside for want of a printed parity	2J, P, 2I, Q3	66	(0, 10, 14)
baryons	TABLE	278 baryons of the PDG table, 14 set aside for want of a printed parity	2J, P, 2I, Q3, strangeness, charm, beauty	184	(0, 10, 40)
baryon_isomulti plet	TABLE	the same 278 baryons, isospin against charge with flavour dropped	2I, Q3	16	(5, 7, 4)
fqh	TABLE	168 quasiparticles of twelve Laughlin states, three of them observed	statistics class, order of the exchange phase, order of the charge, inverse filling fraction	30	(0, 15, 4)
bosonqp	TABLE	15 bosonic excitations -- composites, Goldstone modes and hybrids, every number derived	2J, Q3	5	(7, 4, 2)
readrezayi	TABLE	363 parafermion primaries of eleven Read-Rezayi states, two observed	statistics class, order of the twist, order of the charge, level	78	(0, 19, 11)
spin4	TABLE	10 spin-4 mesons, two of them without a printed mass	P, 2I, Q3	9	(4, 5, 3)
nucbands	TABLE	2,145 nuclear excited states in magnetic and antimagnetic rotational bands; 27 bands and 93 levels refused for carrying no quantum number	2I, parity	121	(2, 63, 2)

2. Five languages and a law

Each of five closure operators takes a finite set of cells to a superset of it. Order and algebra are the sublattice hull under coordinatewise min and max; geometry is hull-completion on coordinate pairs; information is closure under coordinatewise join alone; statistics is 2-determinacy -- the set equals every box point whose two-coordinate projections all occur in it. A language CLOSES an index when applying it returns the index unchanged. The companion paper derives eight clauses relating them; this paper needs only the containments.

contained	in	source
order	algebra	Clause B
algebra	order	Clause B
information	algebra	Clause C
information	order	Clause C
statistics	algebra	6d, three-line proof
statistics	order	6d, three-line proof

contained	in	source
statistics	geometry	Clause G.3

Theorem 1. The lawful channels are the down-sets of that law.

Write $cl[L]$ for the closure of an index X under language L . Every operator is extensive: X is a subset of $cl[L]$ for all L . Suppose (a, b) is one of the seven, so $cl[a]$ is a subset of $cl[b]$, and suppose b closes X , meaning $cl[b] = X$. Then $cl[a]$ is a subset of X , and by extensivity X is a subset of $cl[a]$; hence $cl[a] = X$ and a closes X . So the set of languages closing any index is closed downward under the law. Conversely each down-set is realised, by exhibition. Enumerating the down-sets of the seven relations over five languages gives exactly 8 of the 32 subsets.

channel	languages that close
K0	(none)
K1	information
K2	statistics
K3	geometry+statistics
K4	information+statistics
K5	geometry+information+statistics
K6	algebra+information+order+statistics
K7	algebra+geometry+information+order+statistics

The law leaves exactly two languages free to close alone: information and statistics. Nothing forces either from anything else. That asymmetry is not decoration -- it is what makes K1 and K2 reachable at all, and section 6 shows that the freedom is exercised very differently by the two.

3. The admissible chart

An index is charted by the triple (K, h, w) : its channel, the length of the longest chain in its cell poset (Mirsky), and the size of the largest antichain (Dilworth). The box is the product of the observed coordinate alphabets, never a declared range. Dilworth's theorem gives $|X| \leq h * w$, so the three are not independent and the product box overstates the space; that inequality was checked on every seated index and holds on all of them (0 violations).

4. The index of first-order indexes

Charting each seated index and taking its cell as a member gives a second-order object. It has 23 vertices on 23 distinct cells -- no two seated indexes share a cell -- it closes in nothing, its demand E is 430, and its own cell is $(0, 5, 11)$, which no member occupies.

Applying the chart to itself is a test the object can fail. A coordinate whose distinct values number 90 percent or more of its members separates everything and therefore groups nothing: it is a row identifier wearing a measurement's clothes.

axis	distinct	of	ratio	verdict
K	8	23	0.3478	measurement
height	16	23	0.6957	measurement
width	17	23	0.7391	measurement

No axis is a row label. At eleven vertices two of the three were row labels -- height at 0.909 and width perfectly injective at 1.000 -- and the reading filed with that finding was that each index brings its own height and width, so the two approach injectivity by construction as the object grows. That reading is refuted by the table above: the object grew and

the labels became measurements. The prediction failed because a COARSENING of a seated index does not bring a new height and width; it lands in the part of the poset its parent already occupies. What the argument really showed is that the defect tracks how the vertex set is built, not how large it is.

5. Channel occupancy -- the bound of Theorem 1 is tight

The 23 vertices occupy K0, K1, K2, K3, K4, K5, K6, K7, and EVERY ONE OF THE EIGHT LAWFUL CHANNELS IS OCCUPIED.

channel	languages that close	seated indexes there
K0	(none)	baryons, channels, fqh, gravity, ions, laws, mesons, observed, readrezayi, terms
K1	information	gravity_bound
K2	statistics	fundamental, inversion, madrule, nucbands, probability
K3	geometry+statistics	fibred, nucshell
K4	information+statistics	spin4
K5	geometry+information+statistics	baryon_isomultiplet
K6	algebra+information+order+statistics	madelung_slot
K7	algebra+geometry+information+order+statistics	bosonqp, madelung

Theorem 1 proves that at most 8 of the 32 subsets can occur, and its converse half was proved by EXHIBITION -- constructing a set of cells for each down-set. That is now superseded by something stronger: every down-set is realised by a chart of real atomic, nuclear or particle data whose members carry quantum numbers. The bound is tight, and it is tight from nature rather than from construction. This was not the case when this paper was first drafted, and section 6 is why the last channel was the last.

6. Theorem 2, and why one channel was the last to fall

Statistics is 2-determinacy: X is the set of box points all of whose 2-coordinate projections occur among X's. At arity 2 there is exactly one 2-subset of the coordinates -- the whole of them -- so the projection is the identity and the reconstruction returns X itself. 2-determinacy therefore holds VACUOUSLY at arity 2, for every set whatever, and statistics closes every arity-2 chart. This is a property of the definition and not of any implementation.

Measured over every sub-chart of every seated index:

arity	statistics closes	does not
2	106	0
3	59	66
4	12	96
5	1	62
6	0	21
7	0	3

Corollary. An arity-2 chart cannot occupy K0 or K1.

Its channel contains statistics, and neither K0 nor K1 does. Measured over the same sub-charts, the arity-2 channels observed are K2 (28), K3 (26), K4 (3), K5 (2), K6 (1), K7 (46) -- K0 and K1 occur zero times, as the corollary requires.

THIS IS WHAT MADE K4 THE HARD CHANNEL, and the difficulty was structural rather than accidental. The law protects K5 and K6 from the free pass, because geometry closing forces statistics and so does the order/algebra block; at those channels the statistics bit is earned by law whatever the arity. $K4 = \{\text{information, statistics}\}$ has neither protection -- nothing forces statistics from information -- so it is the only channel above K1 whose extra content is exactly the bit an arity-2 chart is given for free. For a long stretch of this work the only charts reaching K4 were arity 2, where half the channel is a gift, and a census over %d modules (section 10) found no chart of any arity reaching it.

IT IS NOW OCCUPIED, AND OCCUPIED AT ARITY 3. The seated index at K4 is spin4, charted on its 3 coordinates -- an arity at which 2-determinacy is not vacuous and statistics has to be earned. So the channel that the theorem predicted would be hardest is reached, and reached in the way that makes it a measurement rather than a free pass. The explanation above survives as an explanation of the difficulty; it is no longer an account of an absence.

7. Seating an overlapping chart

“They can be seated with overlaps so long as it is not an overlap of same information. An overlap of values in two different languages should tell us two parts of definition contained in that overlapped position. Information is information. But its relative position in this index is information about an object.”

A coarsening -- the same members charted on fewer coordinates -- overlaps its parent totally. The ruling above admits it when it is not the same information. Six readings of that phrase were charted against all proper sub-charts of the seated indexes: 'channel differs from its parent' admits 109, 'cell differs from its parent' 254, 'cell no seated vertex holds' 252, and 'CHANNEL no seated vertex holds' admits 6. The third admits 117 coarsenings of a single index; the fourth is bounded, and it is what the ruling says, since the ruling names languages and the channel is the set of languages that close a chart. Text and arithmetic select the same reading.

Four grounds are tested, and a candidate must clear all four:

- NOVEL CHANNEL -- the chart reaches a channel no seated index reaches.
- NOT A RELABELLING -- it has strictly fewer cells than its parent; a chart that separates exactly as much is the parent renamed.
- REACH STABLE -- the channel does not depend on where the construction stopped: no late arrival, no oscillation, and a majority of reaches.
- COORDINATE FORCED -- the channel survives a faithful re-coordinatisation. Two addresses inducing the identical partition of the identical members are one chart written twice.

Of six candidates, 3 seat and 4 are refused.

chart	channel	cells	verdict / ground failed
gravity (B, F, X)	K1	26	SEATED
madelung (n+l, k)	K6	82	SEATED
baryons (2I, Q3)	K5	16	SEATED
gravity (B, F, X, E)			refused: reach stable
ions (sl, tl)			refused: novel channel, reach stable
madrule (S_a, l_d)			refused: novel channel, reach stable
nucshell (l, sigma)			refused: coordinate forced

8. Two measurements that are about physics

8.1 A horizon threshold visible in the closure algebra

One seated coarsening charts nuclide-charge states on (horizon-bound class, forced angular momentum, spin-decade rank), dropping the spacetime dimension. Its bound class comes from the singly-rotating Myers-Perry horizon condition, which has the Kerr bound in four dimensions, a bound in five, and no bound at all from six upward, where the ultraspinning branch opens. Sweeping the dimension ceiling:

dimensions admitted	channel
D ≤ 5	K7
D ≤ 6	K1
D ≤ 7	K1
D ≤ 8	K1
D ≤ 9	K1
D ≤ 10	K1
D ≤ 11	K1

Read in four and five dimensions the bound structure closes in all five languages. Admit the sixth and four of the five break at once, leaving information alone, and it never moves again. The closure operators are told nothing about dimension; the threshold appears at the dimension the theorem names. The chart is a join-semilattice that is not a lattice -- 0 join counterexamples against 32 meet, in 325 unordered pairs. We record the coincidence and offer no mechanism for why losing a bound should cost four languages rather than three.

8.2 What the periodic table predicts against what it does

Two indexes chart the shell fibration: one on the configuration the Madelung rule predicts, one on the configurations a register of this corpus banks as observed. They are seated together by ruling. The predicted chart has 108 cells at the matched reach and closes statistics; the observed has 98 cells at (0, 20, 13) and closes nothing. 25 of the 108 addresses differ and 20 of the configurations do -- and those 20 are exactly the Madelung exceptions this project indexes separately, so the divergence is not an unknown quantity.

The sharpest difference is one neither chart shows alone. 12 elements -- Cr, Cu, Nb, Ru, Pd, Pr, Tb, Pt, Pa, Pu, Bk, Rf -- LOSE occupancy in a subshell as Z increases by one: a filled subshell gives an electron up. The Madelung prediction is monotone in Z by construction and can never do this, verified at every step of the reach. Neither chart supersedes the other.

9. What the register predicts, and the law that bounds it

Write J(X) for the closure of an index under coordinatewise join and E(X) for the number of cells in J(X) that X does not hold. The corpus reads E as the number of predictions an index can make. Measured over every seated index small enough to close -- one, at 914 cells, is not, and it is named rather than dropped -- the register demands 3,206 cells it does not hold. 16 indexes predict; 6 are complete.

index	E
baryons	1012
readrezayi	678
channels	367
ions	296
laws	254
fibred	140
probability	112
terms	98
fqh	71
observed	56

index	E
fundamental	46
inversion	24
madrule	18
mesons	15
nucshell	14
nucbands	5

The partition is exact -- every $E = 0$ index closes under information and every $E > 0$ index does not -- and it is close to definitional, since E is the join deficit and the information closer is the join closer. It is said so that it is not mistaken for a result. The result is the numbers, and they say the register is overwhelmingly incomplete.

An unadjudicated E is a count of QUESTIONS and not of objects. A demanded cell is one of three things -- FORBIDDEN, a bound rules the combination out; emptiness is a theorem; UNPLACED, the object exists and the source gives it no coordinates; OPEN, physical, placeable, and nothing there -- a real prediction -- and only the third is a prediction. Which of the three a cell is turns out to be settled by a theorem rather than by how much physics one is willing to bring to it.

Theorem 3. The demand invents no coordinate value.

For every coordinate i the projection of $J(X)$ onto i equals the projection of X onto i . PROOF. X is contained in $J(X)$. Conversely $J(X)$ is generated from X by repeated componentwise max, and $\max(a, b)$ is either a or b , so every coordinate of every generated cell is a coordinate value already present in X ; induct on the generation. COROLLARY: no bound constraining a single coordinate can forbid a demanded cell.

Measured on the largest prediction set in the register: of 1012 demanded baryon cells, ZERO carry an even doubled spin -- although $2J$ even is exactly what the spin-statistics of a three-quark state rules out. The strongest single-coordinate fact available about baryons adjudicates nothing, and Theorem 3 says why.

Theorem 4. A monotone bound forbids nothing.

Let B be a set of cells with X contained in B and B closed under componentwise max. Then $J(X)$ is contained in B , because $J(X)$ is the least max-closed set containing X ; so B forbids no demanded cell. In particular a bound of the form $x_i \leq f(x_j, \dots)$, with i not among the j and f non-decreasing in each argument, is max-closed: if a and b lie in B and c is their join, then $c_i = \max(a_i, b_i)$ is one of them, say a_i , and $a_i \leq f(a_j) \leq f(c_j)$ because $a_j \leq c_j$ for each j and f is non-decreasing.

7 bounds were derived for the seated indexes, each from a stated physical law, each checked against every member of its index before being applied to the demand, and none fitted to the demand. 6 of the 7 are monotone, and the monotone ones forbid nothing at all.

index	bound	from	monotone	forbidden
baryons	Gell-Mann--Nishijima : $ 2I_3 \leq 2I$, $2I = 2I_3$ (mod 2)	SU(2) isospin weights + the quark charges	NO	593
fibred	$l \leq n-1$; $k \leq 2(2l+1)$ - 1	radial node count; Pauli exclusion	yes	0
ions	$l \leq n-1$ at both ends; Pauli on k and g ; $q \leq k$	radial node count; Pauli exclusion	yes	0
madrule	$\text{occ} \leq 2(2 \text{ floor}((S_a-1)/2) + 1)$	Pauli exclusion through the hidden l_a	yes	0

index	bound	from	monotone	forbidden
nucshell	$l = 0 \Rightarrow \text{sigma} = +1$	$j = l \pm 1/2$ and $j \geq 0$	yes	0
observed	$l \leq n-1$; $k \leq 2(2l+1) - 1$	radial node count; Pauli exclusion	yes	0
terms	$\text{mult} = 1 \Rightarrow$ the term is not SHORT	Russell-Saunders: a singlet has one level	yes	0

The zeros are not a failed search. $l \leq n-1$ is the radial node count $n - l - 1$ being non-negative; the Pauli caps count the spin-orbitals of a subshell; $q \leq k$ says one cannot ionise more electrons than are present; $l = 0$ forcing the upper spin-orbit branch is $j = l \pm 1/2$ with $j \geq 0$; and a Russell-Saunders singlet has one level, so it cannot fall short of its own multiplet. All are theorems, all hold on every seated member without exception, and by Theorem 4 none of them CAN forbid a demanded cell. The hypothesis is verified rather than read off the algebra: each bound's admissible set is enumerated over its index's own product box and closed under join by exhaustion.

Theorem 5. The quark model forbids no cell of the meson chart.

For a quark-antiquark state $P = (-1)^{(L+1)}$, $C = (-1)^{(L+S)}$, S is 0 or 1, and J runs from $|L - S|$ to $L + S$. CLAIM: every (J, P) with J a non-negative integer and $P = \pm 1$ is realised. PROOF, four cases. $P = -1$ requires L even: for J even take $(L, S) = (J, 0)$; for J odd take $(J - 1, 1)$, which is even and non-negative and has $L + S = J$; $J = 0$ forces $L = S$, and $L = S = 0$ gives $0-$. $P = +1$ requires L odd: for J odd take $(J, 0)$; for J even and at least two take $(J - 1, 1)$; $J = 0$ forces $L = S$, and $L = S = 1$ gives $0+$. Checked by exhaustion to $J = 12$ with no unreached pair.

The exotic quantum numbers $--0--$, $0+-$, $1-+$, $2+-$ are forbidden in J^PC , and C is not a coordinate of this chart. It was refused for TOTALITY: 168 of the 250 mesons in the source carry no C at all, and a chart carries only coordinates all its members have. THE REFUSAL WAS CORRECT AND IT HAS A PRICE, and this is where the price is paid. The meson index has 15 demanded cells and the quark model, fully stated, empties none of them. A coordinate refused for totality is adjudication power given up, and the trade is visible only once someone tries to spend it.

The one bound that forbids

By Theorem 4 only a bound that is antitone somewhere, or carries a congruence, can forbid. One qualifies here and it qualifies twice over. Gell-Mann-Nishijima gives the isospin projection $2^*I_3 = 2Q - Y$ with $Y = B + S + C + B' + T$, and I_3 must be a weight of the isospin- I representation: $|2^*I_3| \leq 2I$, an absolute value, and $2I$ congruent to 2^*I_3 modulo two, a congruence. Both break max-closure. The baryon chart carries Q, S, C, B and I as coordinates, so the bound is expressible on it, and it FORBIDS 593 of the 1012 demanded cells -- 58.6 per cent of the largest prediction set in the register, emptied by theorem rather than by observation.

The relation is not imported. Writing n_a for quarks minus antiquarks of flavour a , $Q = (2/3)(n_u + n_c + n_t) - (1/3)(n_d + n_s + n_b)$, $I_3 = (n_u - n_d)/2$, $S = -n_s$, $C = n_c$, $B' = -n_b$, $T = n_t$ and $B = (\text{sum of } n_a)/3$, whence $Q - I_3 = n_u/6 + n_d/6 + (2/3)(n_c + n_t) - (1/3)(n_s + n_b) = Y/2$ identically. All three legs -- the charge built from the quark charges, the isospin projection, and the identity -- are re-derived from the source's own quark strings: 292 baryon rows, 292 parse, and 292, 292 and 292 clean without exception.

The same check on the mesons found a fault in the source. 191 of 250 meson rows parse -- the rest are flavour mixtures, which is why S, C and B are not coordinates of the meson chart and therefore why the bound is inexpressible there -- and of those the charge and the identity are clean at 191 and 191 while the isospin projection fails at 189. The two failures are one state and its antiparticle, $B(s_2)^*(5840)0$ and $B(s_2)^*(5840)\sim 0$, carrying a strange-beauty content with doubled isospin one, while two other states of the identical content carry zero in the same file. Such a pair holds no up or down quark, so $I_3 = 0$ and $I = 1/2$ has no weight to sit on. IT IS RECORDED AND NOT REPAIRED, and the cost is measured: with the isospin set to zero on both rows the meson index has 66 cells, $E = 15$ and chart position $(0, 10, 14)$ -- identical before and after, because both affected cells are already occupied by other members.

E is a deficit against an operator, and so is forbidding

Every E above is a JOIN deficit. The same cells measured against the ORDER closure give a different deficit and a different verdict. On the 90 drawn positions of the period-by-group periodic layout the join deficit is 0 and the order deficit is 36, and the hydrogenic bound $l \leq n-1$ -- read through the layout's own rule for which subshell sits at which group -- forbids 25 of the order ghosts and 0 of the join ghosts. By Theorem 4 it could not have been otherwise: expressed on (n, l, k) that bound is monotone; expressed on $(\text{period}, \text{group})$ it is not.

Two consequences, and the second is a caution. First, an earlier statement of this project cited that layout's ghosts as the precedent for adjudicating the join deficits tabled above without distinguishing the operators; the three bins survive intact and the precedent is re-attributed rather than withdrawn. Second, FORBIDDING POWER IS A PROPERTY OF THE CHART AND NOT OF THE BOUND: one bound forbids on one coordinatisation of the elements and nothing on two others. The layout on which it has teeth is the one this project WITHDREW as over-representation. A chart that can forbid is not thereby a better chart, and the inference is unsound in both directions.

UNPLACED separated from OPEN

“A demanded cell is UNPLACED if and only if the index's own source holds a row the chart declined to place which supplies all but one of the coordinates and agrees with the cell on every one of them.”

The strength condition is the whole rule. A source row missing one coordinate names a line of cells and pins each of them; a row missing two names a plane and pins nothing. Without it, four thousand unparsed spectroscopic labels would explain every empty cell in one of these charts. Rows below the strength are counted apart and never used. For 10 of the indexes the source has no gap at all, and that is measured rather than assumed: source rows are counted against rows charted and equality is required.

index	E	forbidden	unplaced	open	undecided
baryons	1012	593	8	411	0
readrezayi	678	0	0	678	0
channels	367	0	23	344	0
ions	296	0	0	296	0
laws	254	0	0	254	0
fibred	140	0	0	140	0
probability	112	0	0	112	0
terms	98	0	0	0	98
fqh	71	0	0	71	0
observed	56	0	0	56	0
fundamental	46	0	0	46	0
inversion	24	0	0	24	0
madrule	18	0	0	18	0
mesons	15	0	3	12	0
nucshell	14	0	0	14	0
nucbands	5	0	2	3	0
TOTAL	3206	593	36	2479	98

593 cells are FORBIDDEN and every one of them is a baryon cell. 36 are UNPLACED, in four charts, and the count of CELLS is deliberately kept apart from the count of OBJECTS: in one chart 3 cells are pinned by 2 objects, because a state lacking only its parity pins both parities and will fill exactly one of them. 2,479 are OPEN. 98 are UNDECIDED, and the reason is exact rather than a shrug -- the loader of that chart discards a refused row without banking its term key, so the tree cannot ask whether any term lost ALL of its levels, and such a term would pin a cell at precisely the required

strength. Banking those keys would settle it.

What the OPEN column is and is not. For the 7 indexes carrying a derived bound it is FINAL AGAINST EVERY MONOTONE BOUND, by Theorem 4 -- a proof and not a survey. For the 8 carrying none it is open against nothing at all, and a bound found tomorrow may empty any of them. The two situations are different and are not summed into one adjective. Neither is a count of undiscovered objects.

10. The census

Every chart-shaped accessor in the research tree was charted, with each module's attempt logged so that coverage is measured and not inferred. 232 of 232 modules were attempted; 4 could not be imported and each is named with the reason. 44 charts were found over 31 modules, of which 24 are neither seated nor excused -- and not one of those is a new first-order index. They are members that are not elements, alternate charts of already-seated member sets (all landing in occupied channels), or withdrawn and duplicate charts.

Charts per channel: K0 15, K1 1, K2 10, K3 7, K4 0, K5 1, K6 2, K7 8. Nothing in the tree -- 0 of 44 charts over 231 modules.

Two artifacts of the census are recorded rather than allowed to read as findings: one chart appears unseated because the census keys on (module, accessor) and the seated row reaches it under a second accessor name, and one is a helper written during this work that reproduces an existing index's members.

11. What was refused, and what that establishes

Five adjudications are reported. Four are refusals and one is a retraction. We set them out because a method that only ever accepts has not been tested.

- A candidate reproducing a published four-figure result of this corpus exactly -- 0 join counterexamples and meet counts 2862, 12489, 40887, 110229 at caps 6, 8, 10, 12 -- was REFUSED. Its channel is the same at nine of ten boxes we could hand it, so the channel is a property of the defining predicate and not of any data; it would sit where it sits in a universe with no atoms in it.
- A channel was shown reachable at arity 3, where statistics must be earned, by three charts of an index's own measured quantities. All three were REFUSED: they were found by searching 120 charts for that channel, which is fitting, and a chart selected because it lands somewhere cannot be evidence that it lands there.
- Two further coarsenings were REFUSED on the reach ground -- one oscillating between channels as the element reach grew, one reaching its channel only at the terminal reach, which is the failure mode that withdrew an earlier chart of this project.
- One coarsening was SEATED and then RETRACTED. The same members under an equally faithful address -- and the alternative is the primitive the source actually banks -- land in an occupied channel, so the chart had no novel channel and was never a candidate. Its apparent result was a fact about which name had been written down. The ground that caught it became the fourth test in section 7.

11 statements this project had asserted were measured to be false in the course of the same work and are listed with their corrections in the accompanying state file. Among them: a parent index described as a complete rectangle closing everything for free, which has density 0.2099 -- 170 cells in a box of 810 -- and is not a down-set; an attribution of a closure property to one coordinate when a second restores it equally; and an instrument that DECLARED an index exempt from its own strongest test rather than measuring whether it was. Run properly, that test confirmed the seating and located the physics in a constraint of the hydrogenic spectrum.

12. What is not claimed

- Completeness. The registry's completeness flag is false and stays false. This is the set of first-order indexes found and survived their tests, not a claim to have found them all.
- That occupying all eight channels closes the subject. It makes the bound of Theorem 1 tight and nothing more. It does not say the eight are the right coordinates, that no further index exists, or that any channel is occupied by the best chart of it -- and one of the eight is held by a single index, so its occupancy rests on one seating.
- That the demand E measures progress. It is reported because it is measured. No index here was built to land on a cell the demand wanted, and one that was would be fitted.
- That the second-order object is itself a first-order index. Its members are indexes and carry no quantum numbers; it fails the criterion of section 1 and is excused by name rather than by silence.

Appendix. Reproduction

Every figure above is read from an instrument at build time. Each instrument is stdlib-only and carries a selftest whose fixtures are this corpus's own recorded numbers; the selftest is the first thing to run and the reports are not to be trusted before it passes.

- `python3 registry.py --selftest` -- the criterion, enforced on every row
- `python3 figure.py --selftest` -- the second-order object and its chart
- `python3 overlaprule.py --selftest` -- the ruling, the four grounds, the refusals
- `python3 overlaprule.py --census` -- re-derives the six candidates
- `python3 boxinvariance.py --selftest` -- the refused theorem, and the test run properly
- `python3 observed.py --selftest` -- the observed fibration against the predicted
- `python3 predict.py --selftest` -- the demand per seated index
- `python3 ghosts.py --selftest` -- the seven laws, the bounds, the adjudication
- `python3 state.py --check` -- the state file against every instrument
- `python3 paper/mipaper.py --selftest` -- that no figure in this paper is typed